**Electric Petrol**

# Infrastructure Upgrade Design Report

Midleton Branch Site - Clustered Shared Storage, High Availability & AD Domain Tree Integration

**Assignment 3 - HCT**

Student: Rujuta Ekhande

Student ID: L00187595

Site: Midleton (Site 50)

IP Allocation: 10.50.0.0/16

Domain: Midleton.ads.electric-petrol.ie

Date: April 2026

**CONFIDENTIAL**

## Revision History and Document Management

| Version | Date | Author | Description |
|---|---|---|---|
| **1.0** | 01/04/2026 | L00187595 | Initial draft |
| **1.1** | 02/04/2026 | L00187595 | Server specification and storage design |
| **2.0** | 05/04/2026 | L00187595 | Final report complete |

# Table of Contents

# Executive Summary

This design report presents the technical plan for upgrading the Electric Petrol Midleton branch site (Site 50, IP range 10.50.0.0/16, domain Midleton.ads.electric-petrol.ie). The project addresses three key requirements:

- Add a third virtualization server at the Midleton site and implement Clustered Shared Storage
- Ensure the site continues to operate in the event of a single server failure
- Reengineer AAA to be part of the domain tree for electric-petrol.ie (as Midleton.ads.electric-petrol.ie)

The existing infrastructure at Midleton consists of two Dell R440 servers, each with 256GB RAM in 16GB DIMMs, 2 x 240GB SSD in RAID 1 on a BOSS card, 4 x 1TB HDDs in RAID5, 2 x 1GB/s onboard Ethernet, and 2 x 10GB/s Ethernet in PCI-e slots. Each server runs Windows Server 2019 Datacentre edition with VMs for domain controller (DCx, 4GB), file server (FSx, 8GB), database server with MSSQL (DBx, 24GB), and Ubuntu 2004 Syslog (SLx, 8GB). A Galleon NTS-6002-GPS time server is available via Ethernet.

# 1. Equipment Audit of Existing Midleton Site

In scenario 1, at the Midleton branch site (Site 50), we installed host servers and used virtualization to build a simple, local solution, appropriate to the size of the business. We installed two host servers and used virtualization to build a simple appropriate solution.

## 1.1 Existing Server Hardware Specification

The servers were Dell R440s with:

- 256GB in 16GB DIMMs
- 2 x 240GB SSD in RAID 1 on a BOSS card for the operating system
- 4 x 1TB HDDs in RAID5 for data
- 2 x 1GB/s on board Ethernet cards
- 2 x 10GB/s Ethernet cards in a PCI-e slot

| • Component | • Specification |
|---|---|
| • **Memory** | • 256GB in 16GB DIMMs |
| • **OS Storage** | • 2 x 240GB SSD in RAID 1 on a BOSS card |
| • **Data Storage** | • 4 x 1TB HDDs in RAID5 |
| • **Onboard Networking** | • 2 x 1GB/s on board Ethernet cards |
| • **Additional Networking** | • 2 x 10GB/s Ethernet cards in a PCI-e slot |
| • **Time Server** | • Galleon NTS-6002-GPS available via Ethernet |

## 1.2 Existing Virtual Machine Configuration

Each host server was configured with Windows 2019 Server (Datacentre edition) and has the following VMs:

| VM Name | Operating System | Role | RAM |
|---|---|---|---|
| **DCx** | Windows 2019 Server | Domain Controller | 4GB |
| **FSx** | Windows 2019 Server | File Server | 8GB |
| **DBx** | Windows 2019 Server | Database Server (MSSQL) | 24GB |
| **SLx** | Ubuntu 2004 | Syslog Server | 8GB |

Total RAM utilised per host: 44GB out of 256GB available (17.2% utilisation).

There is a Galleon NTS-6002-GPS time server available via Ethernet.

## 1.3 Midleton Site Network Allocation

The Midleton branch site (Site 50) has been allocated the IP range 10.50.0.0/16 with the following VLAN structure:

| VLAN | Purpose | Network |
|---|---|---|
| **1** | ISP | 10.50.1.0/24 |
| **2** | Hosts | 10.50.2.0/24 |
| **3** | Staff Radio | 10.50.3.0/24 |
| **4** | Power Infrastructure | 10.50.4.0/24 |
| **5** | Technical Clients | 10.50.5.0/24 |
| **6** | Infrastructure Servers | 10.50.6.0/24 |
| **7** | Telephony | 10.50.7.0/24 |
| **8** | HVAC Infrastructure | 10.50.8.0/24 |
| **9** | Storage | 10.50.9.0/24 |
| **10** | NetMan | 10.50.10.0/24 |
| **11** | Security Cameras | 10.50.11.0/24 |
| **12** | Server OOBM | 10.50.12.0/24 |

## 1.4 Midleton Site IP Address Assignments

The following fixed IP addresses are assigned for key infrastructure on the Midleton site:

| Device | VLAN | IP Address |
|---|---|---|
| **Host Server 1 (Host11)** | VLAN 2 - Hosts | 10.50.2.21 |
| **Host Server 2 (Host12)** | VLAN 2 - Hosts | 10.50.2.22 |
| **Domain Controller 1 (DC1)** | VLAN 6 - Infra Servers | 10.50.6.31 |
| **Domain Controller 2 (DC2)** | VLAN 6 - Infra Servers | 10.50.6.32 |
| **Storage Server 1** | VLAN 6 - Infra Servers | 10.50.6.41 |
| **Storage Server 2** | VLAN 6 - Infra Servers | 10.50.6.42 |
| **Database Server 1 (DB1)** | VLAN 6 - Infra Servers | 10.50.6.51 |
| **Database Server 2 (DB2)** | VLAN 6 - Infra Servers | 10.50.6.52 |
| **Storage 1 (iSCSI)** | VLAN 9 - Storage | 10.50.9.41 |
| **Storage 2 (iSCSI)** | VLAN 9 - Storage | 10.50.9.42 |

# 2. Technical Tasks Required for the Upgrade

This section details the exact technical tasks for upgrading the Midleton branch site (Site 50, 10.50.0.0/16).

## 2.1 Work Packages Overview

| WP | Task | Description |
|---|---|---|
| **WP1** | Server Procurement | Procure and install Dell PowerEdge R450 (third server) at Midleton site |
| **WP2** | Storage Deployment | Deploy Dell PowerVault ME5024 shared storage array at Midleton |
| **WP3** | Existing Server Upgrades | Add HBA controllers and reconfigure 10GbE NICs for iSCSI on both Dell R440s |
| **WP4** | Cluster Configuration | Configure VMware HA, DRS, and Clustered Shared Volumes |
| **WP5** | VM Migration | Migrate all VMs from local to shared storage using Storage vMotion |
| **WP6** | AD Restructuring | Migrate standalone AD to Midleton.ads.electric-petrol.ie child domain |
| **WP7** | Testing | Validate failover, authentication, and performance |

## 2.2 Implications and Considerations

Downtime: Each existing Dell R440 server (Host11 at 10.50.2.21, Host12 at 10.50.2.22) requires a 2-4 hour maintenance window for HBA controller installation. VM migration via Storage vMotion is non-disruptive.

Licensing: Windows Server 2019 Datacentre edition licenses cover unlimited VMs per host. VMware vSphere Enterprise Plus licensing is required for HA and DRS.

Network: The existing 2 x 10GB/s Ethernet cards in each PCI-e slot will be dedicated to iSCSI storage traffic on VLAN 9 (Storage, 10.50.9.0/24). Storage IPs: 10.50.9.41 and 10.50.9.42.

Backup: Full backups of all VMs (DCx 4GB, FSx 8GB, DBx 24GB MSSQL, SLx 8GB Ubuntu 2004) must be taken before any hardware changes.

# 3. Upgrades for Existing Servers

Both existing Dell R440 servers at Midleton (Host11 at 10.50.2.21 and Host12 at 10.50.2.22), each with 256GB RAM, 2 x 240GB SSD BOSS, 4 x 1TB HDD RAID5, 2 x 1GB/s + 2 x 10GB/s Ethernet, require the following upgrades:

## 3.1 Hardware Upgrades

| Component | Current (Dell R440) | Change Required | Justification |
| --- | --- | --- | --- |
| **Storage HBA** | None (onboard PERC) | Add Dell HBA330 SAS/SATA HBA | Shared SAN/iSCSI storage connectivity |
| **10GbE NICs** | 2 x 10GB/s PCI-e (general) | Dedicate for iSCSI with MPIO on VLAN 9 (10.50.9.0/24) | Isolated high-speed storage network |
| **Memory** | 256GB in 16GB DIMMs | No change | Only 44GB of 256GB used per host |
| **Local Storage** | 4 x 1TB HDDs in RAID5 | Retain for scratch/ISOs | VM storage moves to shared SAN |
| **OS Storage** | 2 x 240GB SSD RAID 1 (BOSS) | No change | BOSS card boot config unchanged |

## 3.2 Software Upgrades

- Update VMware ESXi to latest 7.0 U3 patch for HA compatibility
- Configure iSCSI Software Adapter on VLAN 9 (Storage, 10.50.9.0/24)
- Add Storage VLAN 9 with MTU 9000 on virtual switches
- Reconfigure NTP to sync with Galleon NTS-6002-GPS time server
- Rejoin ESXi hosts to Midleton.ads.electric-petrol.ie domain

# 4. Specification of the New Third Server

The third server for the Midleton site must be compatible with the existing Dell R440 infrastructure. The Dell PowerEdge R450 is the recommended successor.

## 4.1 Dell PowerEdge R450 Specification

| Component | Specification | Notes |
| --- | --- | --- |
| **Model** | Dell PowerEdge R450 (1U Rack) | Direct successor to Dell R440 |
| **Processor** | 2 x Intel Xeon Silver 4314 (16C/32T) | Matches/exceeds existing R440 |
| **Memory** | 256GB DDR4 in 16GB DIMMs | Matches existing (256GB in 16GB DIMMs) |
| **OS Storage** | 2 x 480GB SSD RAID 1 on BOSS-S2 | Upgraded from 2 x 240GB SSD BOSS |
| **Local Data Storage** | 4 x 1TB SAS HDDs in RAID5 | Matches existing (4 x 1TB HDDs RAID5) |
| **Storage HBA** | Dell HBA355i SAS/SATA | Pre-installed for shared SAN/iSCSI |
| **Onboard Networking** | 2 x 1GB/s Ethernet | Matches existing (2 x 1GB/s onboard) |
| **Additional Networking** | 2 x 10GB/s Ethernet PCI-e | Matches existing; dedicated to iSCSI on VLAN 9 |
| **Power Supply** | 2 x 600W redundant hot-swap PSU | Hardware fault tolerance |

## 4.2 New Server Network Configuration

The third server (Host13) will be assigned the following IPs on the Midleton network (10.50.0.0/16):

| Interface | VLAN | IP Address |
| --- | --- | --- |
| **Management NIC 1** | VLAN 2 - Hosts | 10.50.2.23 |
| **Management NIC 2** | VLAN 10 - NetMan | 10.50.10.23 |
| **iSCSI NIC 1** | VLAN 9 - Storage | 10.50.9.43 |
| **iSCSI NIC 2** | VLAN 9 - Storage | 10.50.9.44 |

## 4.3 VM Distribution Across Three Hosts

With three servers, VMs are distributed for redundancy with anti-affinity rules:

| Host | VMs | RAM Used | Key IPs (VLAN 6) |
| --- | --- | --- | --- |
| **Host11 (Dell R440) - 10.50.2.21** | DC1 (4GB), FS1 (8GB) | 12GB/256GB | DC1: 10.50.6.31 |
| **Host12 (Dell R440) - 10.50.2.22** | DB1 (24GB), SL1 (8GB) | 32GB/256GB | DB1: 10.50.6.51 |
| **Host13 (Dell R450) - 10.50.2.23** | DC2 (4GB), FS2 (8GB) | 12GB/256GB | DC2: 10.50.6.32, DB2: 10.50.6.52 |
| **Failover** | Any host absorbs full workload | 88GB max/256GB | N+1 redundancy assured |

# 5. Clustered Shared Storage Implementation

Clustered Shared Volumes (CSV) provide shared storage accessible by all three ESXi hosts at the Midleton site simultaneously, enabling VM high availability and live migration.

## 5.1 Shared Storage Array

| Component | Specification |
|---|---|
| **Storage Array** | Dell PowerVault ME5024 (2U, 24-bay SFF) |
| **Drives** | 12 x 2.4TB 10K SAS HDDs in RAID 10 (~12TB usable) |
| **Cache** | 2 x 4GB read/write cache per controller |
| **Connectivity** | 4 x 10GbE iSCSI (2 per controller, dual controllers) |
| **RAID** | RAID 10 for performance and redundancy |
| **Protocol** | iSCSI over 10GbE |

## 5.2 Storage Network Design - Midleton VLAN 9

The dedicated Storage VLAN 9 (10.50.9.0/24) isolates iSCSI traffic at the Midleton site. Each host's 2 x 10GB/s Ethernet PCI-e cards are configured for iSCSI multipathing:

| Device | VLAN 9 IP (10.50.9.x) | Role |
|---|---|---|
| **Host11 (Dell R440) - Path A** | 10.50.9.21 | iSCSI Initiator Path 1 |
| **Host11 (Dell R440) - Path B** | 10.50.9.22 | iSCSI Initiator Path 2 |
| **Host12 (Dell R440) - Path A** | 10.50.9.23 | iSCSI Initiator Path 1 |
| **Host12 (Dell R440) - Path B** | 10.50.9.24 | iSCSI Initiator Path 2 |
| **Host13 (Dell R450) - Path A** | 10.50.9.43 | iSCSI Initiator Path 1 |
| **Host13 (Dell R450) - Path B** | 10.50.9.44 | iSCSI Initiator Path 2 |
| **Storage Array Controller A/B** | 10.50.9.41 / 10.50.9.42 | iSCSI Target (dual controller) |

All iSCSI ports use MTU 9000 (jumbo frames) with Round Robin multipathing policy.

## 5.3 VMFS Datastore Layout

| Datastore Name | Size | Purpose |
|---|---|---|
| **Midleton-DS-Production** | 6TB | Production VMs (DC, FS, DB, Syslog) |
| **Midleton-DS-Templates** | 2TB | VM templates, ISOs, deployment images |
| **Midleton-DS-Backup** | 4TB | VM snapshots and backup storage |

## 5.4 VM Migration to Shared Storage

Existing VMs are migrated from local datastores (4 x 1TB HDDs RAID5 per R440) to shared storage using Storage vMotion (non-disruptive):

- Phase 1: Migrate DC1, DC2 (Windows 2019, 4GB each) - smallest, lowest risk
- Phase 2: Migrate SL1 (Ubuntu 2004 Syslog, 8GB)
- Phase 3: Migrate FS1, FS2 (Windows 2019 File Server, 8GB each)
- Phase 4: Migrate DB1 (Windows 2019 MSSQL, 24GB) during maintenance window
- Phase 5: Verify all VMs on shared storage; reclaim local datastores

# 6. High Availability Design

The Midleton site must continue operating during a single server failure. VMware vSphere HA combined with Clustered Shared Storage on VLAN 9 (10.50.9.0/24) achieves this.

## 6.1 vSphere HA Configuration

- Admission Control: Reserve resources for 1 host failure (33% reservation)
- Host Monitoring: Heartbeat datastores on Midleton shared storage
- VM Restart Priority: High for DC1/DC2 and DB1, Medium for FS1/FS2, Low for SL1
- VM Monitoring: Heartbeat via VMware Tools
- Host Isolation Response: Power off and restart VMs on surviving hosts
- Network Redundancy: Management on dual 1GB/s NICs with NIC teaming on VLAN 2 (10.50.2.0/24)

## 6.2 DRS Configuration

- Automation Level: Fully Automated
- Migration Threshold: Level 3 (moderate)
- Anti-Affinity: DC1 (10.50.6.31) and DC2 (10.50.6.32) never on same host
- Anti-Affinity: DB1 (10.50.6.51) and DB2 (10.50.6.52) never on same host
- Live migration via vMotion over VLAN 2 (10.50.2.0/24) for zero-downtime rebalancing

## 6.3 Failover Capacity Analysis

| Scenario | Available RAM | Status |
|---|---|---|
| **Normal (3 hosts at Midleton)** | 768GB total, 88GB used (11.5%) | Fully operational with DRS |
| **Single Host Failure (2 hosts)** | 512GB total, 88GB used (17.2%) | All VMs operational via HA |
| **Two Host Failure (1 host)** | 256GB total, 88GB used (34.4%) | All VMs on single host (degraded) |

Each host has 256GB RAM (in 16GB DIMMs). With only 44GB utilised per VM set, any single host can absorb the full workload (VMware, 2024).

# 7. New Active Directory Domain Tree Design

The Midleton branch currently operates a standalone AD domain. It must be reengineered to join the electric-petrol.ie domain tree as Midleton.ads.electric-petrol.ie.

## 7.1 Current AAA Configuration

- Windows 2019 domain controller (DCx) using 4GB RAM - standalone domain
- ESXi hosts (10.50.2.21, 10.50.2.22) joined to local AD
- vCenter Server joined to local AD for VM management
- Ubuntu 2004 Syslog server (SLx, 8GB) collecting security audit logs

## 7.2 Target Domain: Midleton.ads.electric-petrol.ie

**Domain Tree Structure:**

electric-petrol.ie (Forest Root - HQ, 10.0.0.0/16)

|-- ads.electric-petrol.ie (AD Services)

|-- Midleton.ads.electric-petrol.ie (Midleton Branch, 10.50.0.0/16)

- Forest Root: electric-petrol.ie at HQ (10.0.0.0/16)
- Child Domain: Midleton.ads.electric-petrol.ie (10.50.0.0/16)
- Trusts: Automatic two-way transitive trusts
- Global Catalog: DC1 (10.50.6.31) and DC2 (10.50.6.32) both as GC servers
- DNS: AD-integrated DNS for Midleton.ads.electric-petrol.ie zone
- Time: PDC Emulator syncs to Galleon NTS-6002-GPS; other DCs sync from PDC

## 7.3 Domain Controller Deployment

| DC | Host / IP | Roles | RAM |
|---|---|---|---|
| **DC1** | Host11 (10.50.2.21) / DC IP: 10.50.6.31 | AD DS, DNS, GC, FSMO roles | 4GB |
| **DC2** | Host13 (10.50.2.23) / DC IP: 10.50.6.32 | AD DS, DNS, GC (replica) | 4GB |
| **Anti-Affinity** | DRS Enforced | DC1 and DC2 never on same host | - |

## 7.4 Migration Steps

1.Deploy forest root electric-petrol.ie at HQ (10.0.0.0/16)

2. Establish VPN/WAN between Midleton (10.50.0.0/16) and HQ

3. Promote DC1 (10.50.6.31) as first child DC for Midleton.ads.electric-petrol.ie

4. Promote DC2 (10.50.6.32) as replica child DC on Host13

5. Migrate users/groups/computers via ADMT from standalone to Midleton.ads.electric-petrol.ie

6. Update Group Policy Objects for new domain
7. Rejoin ESXi Host11 (10.50.2.21), Host12 (10.50.2.22), Host13 (10.50.2.23) to new domain
8. Rejoin vCSA to Midleton.ads.electric-petrol.ie
9. Configure Syslog SLx (Ubuntu 2004, 8GB) to receive audit logs from new domain
10. Decommission old standalone DC after validation

## 7.5 Security Policies

- Password Policy: 12+ characters, complexity, 90-day rotation
- Account Lockout: 5 failed attempts, 30-minute lockout
- Kerberos: Domain tree trusts enable single sign-on across electric-petrol.ie
- Audit: Events forwarded to Ubuntu 2004 Syslog server (SLx, 8GB) at Midleton
- RADIUS/NPS: Centralised RADIUS on DCs for network equipment auth

# 8. Implementation Management Plan

This section outlines the phased implementation plan for the Midleton site (10.50.0.0/16) infrastructure upgrade.

## 8.1 Phase 1: Hardware Deployment

| Task | Details | Duration |
|---|---|---|
| **Rack and cable Host13 (Dell R450)** | Install in Midleton server room, connect 2x1GB/s + 2x10GB/s | 1 day |
| **Install Dell PowerVault ME5024** | Rack storage array, cable iSCSI to VLAN 9 (10.50.9.0/24) | 1 day |
| **Configure iSCSI network** | Storage1: 10.50.9.41, Storage2: 10.50.9.42, host paths: .21-.24,.43-.44 | 1 day |
| **Verify Galleon NTS-6002-GPS** | Confirm NTP via Ethernet to all hosts | 0.5 day |

## 8.2 Phase 2: VMware Configuration

| Task | Details | Duration |
|---|---|---|
| **Install ESXi 7.0 on Host13** | Configure mgmt IP 10.50.2.23 on VLAN 2 | 0.5 day |
| **Add Host13 to vCenter cluster** | Join Midleton-Cluster, configure DRS/HA | 0.5 day |
| **Create shared datastores** | Midleton-DS-Production (6TB), Templates (2TB), Backup (4TB) | 1 day |
| **Storage vMotion all VMs** | Migrate DC1, DC2, SL1, FS1, FS2, DB1 to shared storage | 2 days |
| **Configure HA/DRS rules** | Anti-affinity: DC1/DC2, DB1/DB2; admission control 33% | 0.5 day |

## 8.3 Phase 3: Active Directory Reengineering

| Task | Details | Duration |
|---|---|---|
| **Establish WAN to HQ** | VPN between Midleton 10.50.0.0/16 and HQ 10.0.0.0/16 | 1 day |
| **Promote DC1 to child domain** | Midleton.ads.electric-petrol.ie, DC1 at 10.50.6.31 | 1 day |
| **Promote DC2 as replica** | DC2 at 10.50.6.32 on Host13, GC enabled | 0.5 day |
| **Migrate users/groups via ADMT** | All accounts to Midleton.ads.electric-petrol.ie | 2 days |
| **Rejoin ESXi hosts to new domain** | Host11 (10.50.2.21), Host12 (10.50.2.22), Host13 (10.50.2.23) | 1 day |

| | | |
|---|---|---|
| **Update vCSA and Syslog config** | Rejoin vCSA; reconfigure SLx for new domain events | 0.5 day |

## 8.4 Phase 4: Testing and Validation

| Task | Details | Duration |
|---|---|---|
| **HA failover test** | Simulate host failure, verify VM restart on surviving hosts | 0.5 day |
| **DRS rebalancing test** | Verify anti-affinity and live migration | 0.5 day |
| **AD replication test** | Verify DC1-DC2 replication, DNS resolution for Midleton.ads.electric-petrol.ie | 0.5 day |
| **Storage failover test** | Simulate iSCSI path failure, verify multipath failover | 0.5 day |
| **Full site failover** | Power off one host, confirm all services operational | 1 day |

# 9. Risk Assessment

The following risk assessment identifies key risks to the Midleton infrastructure upgrade project.

| Risk | Likelihood | Impact | Mitigation | Residual Risk |
|---|---|---|---|---|
| **Single host failure during migration** | Medium | High | Schedule Storage vMotion in maintenance window; keep local copies until verified | Low |
| **iSCSI storage network failure** | Low | Critical | Dual controller (10.50.9.41/42), multipath with 2 paths per host, jumbo frames | Low |
| **AD migration data loss** | Low | High | Full AD backup before migration; ADMT rollback capability | Low |
| **DNS resolution failure post-migration** | Medium | High | Keep old DNS records during transition; conditional forwarders to HQ | Low |
| **WAN link failure to HQ** | Medium | Medium | Midleton DCs cache credentials locally; site operates autonomously | Low |
| **Hardware delivery delay for Host13** | Medium | Medium | Order Dell R450 early; use temporary hardware if needed | Medium |

# 10. Conclusion

## 10.1 Summary of Design Achievements

This report has presented a comprehensive infrastructure upgrade plan for the Electric Petrol Midleton branch site (Site 50), addressing all three customer requirements within the constraints of the existing two-server environment. The design introduces a third Dell R450 virtualisation host (Host13 at 10.50.2.23), Dell PowerVault ME5024 shared SAN storage on a dedicated VLAN 9 (10.50.9.0/24), and a complete Active Directory restructure from a standalone domain into the Midleton.ads.electric-petrol.ie child domain within the electric-petrol.ie forest. These changes transform the Midleton site from an isolated, single-point-of-failure deployment into a resilient, centrally managed branch that aligns with enterprise-grade infrastructure standards.

## 10.2 Critical Evaluation of Virtualisation Strategy

The decision to deploy VMware vSphere with High Availability (HA) and Distributed Resource Scheduler (DRS) is grounded in well-established virtualisation theory. According to the VMware vSphere 7.0 Resource Management Guide (VMware, 2024), server consolidation through virtualisation typically achieves consolidation ratios of 10:1 to 20:1 in branch office scenarios, reducing both capital and operational expenditure. In this design, the three-host cluster with 768 GB total RAM and only 88 GB utilised represents a consolidation ratio well within industry norms, while providing substantial headroom for future growth. The VMware vSphere HA admission control policy configured at 33% (one host equivalent) ensures that in the event of any single host failure, the remaining two hosts can absorb all workloads without performance degradation. This aligns with the N+1 redundancy model described in the VMware vSphere Availability documentation (VMware, 2024), which is considered the minimum acceptable standard for business-critical branch infrastructure. The use of DRS further optimises resource allocation by dynamically migrating VMs across hosts based on CPU and memory utilisation thresholds, a capability identified as essential for maintaining consistent service levels in multi-host environments.

An alternative approach would have been to adopt VMware vSAN, which eliminates the need for external shared storage by aggregating local disks across hosts into a distributed datastore. However, vSAN requires a minimum of three hosts with flash caching devices and introduces additional licensing costs. Given that the existing Dell R440 servers use BOSS cards for OS and RAID5 HDDs for data, retrofitting them for vSAN would require significant hardware investment. The PowerVault ME5024 SAN approach was therefore selected as more cost-effective for a three-host branch deployment, consistent with Dell Technologies guidelines for SMB storage consolidation (Dell Technologies, 2024).

## 10.3 Storage Architecture and Clustered Shared Volumes

The implementation of Clustered Shared Volumes (CSV) on the PowerVault ME5024 represents a significant advancement over the previous per-host local storage model. CSV enables multiple cluster nodes to simultaneously access the same NTFS or ReFS volumes, which is a prerequisite for live migration (vMotion) and failover clustering. The RAID configuration selected (RAID 10 for performance-critical VMs, RAID 5 for general storage) follows tiered storage methodology where write-intensive workloads such as MSSQL databases benefit from RAID 10's superior write performance, while read-heavy workloads such as file services can tolerate RAID 5's lower cost per gigabyte. The dedicated iSCSI network on VLAN 9 (10.50.9.0/24) with 10 Gbps connectivity ensures that storage traffic is isolated from production VM traffic, preventing bandwidth contention. This network segmentation approach aligns with best practices outlined in the VMware vSphere Networking Guide (VMware, 2024), which recommends dedicated storage networks to maintain consistent I/O performance and reduce latency below the 2ms threshold required for real-time transaction processing.

## 10.4 Active Directory Reengineering and Security Implications

The reengineering of the Midleton AAA infrastructure from a standalone domain to a child domain (Midleton.ads.electric-petrol.ie) within the electric-petrol.ie forest is arguably the most architecturally significant change in this design. As documented in the Microsoft Active Directory Domain Services overview (Microsoft, 2024), forest and domain design decisions have long-term implications for security boundaries, administrative delegation, and replication topology. By adopting a child domain model with two-way transitive trusts, the Midleton site gains several critical capabilities: centralised Group Policy management from the forest root, Kerberos-based authentication across all Electric Petrol sites, and a unified Global Catalogue for resource discovery. The deployment of dual domain controllers (DC1 at 10.50.6.31, DC2 at 10.50.6.32) with Active Directory Sites and Services configured for intra-site replication provides fault tolerance for authentication services. Microsoft's documentation explicitly states that a site should never rely on a single domain controller, as the loss of all DCs in a site renders all domain-joined resources inaccessible until replication from a remote DC can be established (Microsoft, 2024). The dual-DC design ensures continuous authentication even during planned maintenance windows. From a security perspective, the integration into the corporate forest enables centralised audit logging and security monitoring through Windows Event Forwarding, which is essential for compliance with data protection regulations including GDPR, particularly relevant for Electric Petrol's operations across EU member states. The DFS Replication service (Microsoft, 2024 DFS) provides efficient file data synchronisation between branch sites and the central data centre using Remote Differential Compression, reducing WAN bandwidth consumption by 60-90% compared to full-file transfers.

## 10.5 High Availability and Fault Tolerance Analysis

The requirement for single-server-failure tolerance has been addressed through multiple layers of redundancy. At the compute layer, the three-host VMware cluster with HA admission control ensures automatic VM restart on surviving hosts within 30-120 seconds of a host failure, depending on VM size and storage latency (VMware, 2024). At the storage layer, the PowerVault ME5024 with dual controllers and redundant power supplies eliminates storage as a single point of failure (Dell Technologies, 2024). At the network layer, NIC teaming on each host provides link-level redundancy. This defence-in-depth approach aligns with the NIST SP 800-53 framework for availability controls (NIST, 2020), which recommends that critical systems implement redundancy at every architectural tier. The Mean Time Between Failures (MTBF) for Dell PowerEdge servers is typically rated at 40,000-60,000 hours, which in a three-server cluster translates to a statistically low probability of concurrent dual-host failure. However, it is important to acknowledge that this design does not protect against site-level disasters such as fire, flooding, or extended power outages. For complete business continuity, a future phase should consider asynchronous replication of critical VMs to DC1 in Mayo, leveraging the infrastructure designed in the A4 component of this project.

## 10.6 Limitations and Future Considerations

While this design meets all stated requirements, several limitations should be acknowledged for transparency. First, the design assumes WAN connectivity to DC1 for AD replication but does not specify WAN bandwidth requirements or failover mechanisms; future work should quantify replication traffic volumes using Active Directory Replication Status Tool metrics. Second, the current VM workload of 88 GB across 8 VMs leaves substantial unused capacity (680 GB), which while beneficial for fault tolerance, represents a capital investment that may not be fully utilised for several years. Third, the design does not address backup and recovery beyond the HA failover capability; a dedicated backup solution such as Veeam Backup and Replication, integrated with the PowerVault storage snapshots, should be considered for comprehensive data protection. Finally, as Electric Petrol expands its EV charging infrastructure across Ireland, the Midleton site may need to support edge computing workloads for real-time charge point management. The three-host cluster provides a foundation for such expansion, and the International Energy Agency's Global EV Outlook (IEA, 2023) highlights the growing importance of intelligent infrastructure management for EV charging networks.

## 10.7 Concluding Remarks

In summary, the proposed infrastructure upgrade transforms the Electric Petrol Midleton site from a vulnerable two-server standalone deployment into a resilient, enterprise-integrated branch infrastructure. The design balances cost-effectiveness with enterprise requirements by leveraging proven technologies (VMware vSphere HA, Dell PowerVault CSV, Active Directory child domains) rather than adopting emerging but less mature alternatives. The three-host cluster with shared storage and dual domain controllers delivers the fault tolerance required by the customer while providing a scalable foundation for Electric Petrol's evolving business needs, including EV charging expansion and smart energy management across its nationwide network of service stations **(Dell Technologies, 2024; VMware, 2024; Microsoft, 2024).**

# Appendix A: Equipment Audit Summary

**Source: Equipment Audit of Existing Sites (Electric Petrol)**

**Physical Server Specification (Dell R440):**

- RAM: 256 GB in 16 GB DIMMs
- Boot: 2 x 240 GB SSD in a RAID 1 BOSS configuration
- Data: 4 x 1 TB HDD in a RAID 5 configuration
- Network: 2 x 1 GB/s onboard Ethernet
- Network: 2 x 10 GB/s PCI-e Ethernet
- Hypervisor: VMware ESXi (version audited)

**Virtual Machine Inventory per Host:**

| VM | OS | RAM | Role |
|---|---|---|---|
| **DCx** | Windows 2019 | 4GB | Domain Controller |
| **FSx** | Windows 2019 | 8GB | File Server |
| **DBx** | Windows 2019 + MSSQL | 24GB | Database Server |
| **SLx** | Ubuntu 2004 | 8GB | Syslog Server |

**Time Server:**

- Galleon NTS-6002-GPS with Ethernet connection
- Provides NTP to all hosts and VMs at the Midleton site

# Appendix B: References


- Dell Technologies (2024) PowerEdge R440 Technical Guide. Available at: https://www.dell.com/support/home/en-ie/product-support/product/poweredge-r440/docs .

- Dell Technologies (2024) PowerEdge R450 Technical Guide. Available at: https://www.dell.com/support/home/en-ie/product-support/product/poweredge-r450/docs .

- Dell Technologies (2024) PowerVault ME5024 Storage Array Specifications. Available at: https://www.dell.com/support/home/en-ie/product-support/product/powervault-me5024/docs .

- IEA (2023) Global EV Outlook 2023. Paris: International Energy Agency. Available at: https://www.iea.org/reports/global-ev-outlook-2023 .

- Microsoft (2024) Active Directory Domain Services Overview. Available at: https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview .

- Microsoft (2024) DFS Replication Overview. Available at: https://learn.microsoft.com/en-us/windows-server/storage/dfs-replication/dfsr-overview .
- NIST (2020) Security and Privacy Controls for Information Systems and Organizations. NIST SP 800-53 Rev 5. Available at: https://csrc.nist.gov/publications/detail/sp/800-53/rev-5/final .
- VMware (2024) vSphere 7.0 High Availability and DRS Administration Guide. Available at: https://docs.vmware.com/en/VMware-vSphere/7.0/vsphere-availability/GUID-33A65FF7-DA22-4DC5-8B18-5A7F97CCA536.html .
- VMware (2024) vSphere 7.0 Networking Guide. Available at: https://docs.vmware.com/en/VMware-vSphere/7.0/vsphere-networking/GUID-35B40B0B-0C13-43B2-BC85-18C9C91BE2D4.html .
- VMware (2024) vSphere 7.0 Resource Management Guide. Available at: https://docs.vmware.com/en/VMware-vSphere/7.0/vsphere-resource-management/GUID-98BD5A8A-260A-494F-BAAE-74781F5C4B87.html .
- VMware (2024) vSphere 7.0 Storage vMotion Documentation. Available at: https://docs.vmware.com/en/VMware-vSphere/7.0/vsphere-vcenter-esxi-management/GUID-AB266895-BAA4-4BF3-894E-47F99DC7B77F.html .